

Matheus Calegari

Software Engineer

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SUMMARY

Software Engineer with experience in full-stack development and data science, working with NestJS, Angular, React, and Python. Background in computer vision, LLMs, and RAG architecture — with projects registered at the INPI (National Institute of Industrial Property) and adopted institutionally. I deliver solutions driven by measurable impact in production environments.

EXPERIENCE

Grão Direto

Software Engineer

Mar 2026 - Present

- Developed and delivered two complete projects from scratch, automating 5 workflows that previously relied entirely on email and manual inspection — using clean architecture, serverless functions, real-time updates, and event-driven flows.
- Established observability and code review practices as team standards, reducing regressions and ensuring preventive maintenance of solutions in NestJS, Angular, and MariaDB.
- Resolved critical production incidents and led improvements in the CI/CD pipeline, increasing delivery reliability.

AeC

Data Scientist (Intern)

Nov 2024 - Feb 2026

- Trained and optimized YOLO models through hyperparameter tuning and compression techniques, achieving 98% accuracy in the highest-frequency classes of the production dataset.
- Developed unit tests that increased code coverage by 30%, reducing the incidence of critical bugs in production.
- Built tools to process and evaluate over 300 KPIs using LLMs.

Robbyson

Full Stack Developer (Intern)

Apr 2024 - Oct 2024

- Developed and implemented new front-end components with React and TypeScript, enhancing usability and user experience.
- Identified and fixed performance bottlenecks in the front-end, back-end, and database, optimizing business logic.
- Implemented WAI-ARIA and WAI-WCAG 2.1 accessibility standards, promoting inclusion for users with disabilities.

EJ. BugWare

Mobile Developer | Tech Lead

Jul 2023 - Jul 2025

- Led the development of a mobile application for the Public Prosecutor's Office of Minas Gerais (MPMG) and the Environmental Military Police Battalion to conduct environmental inspections, coordinating a team of 3 developers in the Flutter framework.
- Built complete CI/CD pipelines, integrated with Firebase Distribution and GitHub Actions, for automated deployment and issue prevention with a test-driven culture.

PROJECTS

Sales Analytics Dashboard — LM Sistemas (Freelance)

2025

Vite · Tanstack Router · Tanstack Query · TypeScript · ShadCN

- Built a comprehensive dashboard, delivering real-time visibility of 9 operational KPIs and enabling data-driven preventive decisions — replacing dispersed manual tracking.
- Identified and fixed inconsistencies in the C#/.NET API during integration, ensuring the integrity of the displayed data.

RAG Architecture Chatbot — IFTM

Apr 2025 - Dec 2025

Python · LLMs open-source · Qdrant · RAG

- Developed a chatbot for accessing admissions process information, achieving 82% Answer Correctness and 98% Hit Rate using exclusively open-source tools.
- Solution adopted by the IFTM Admissions Coordination to support 30+ annual selection processes, with deployment scheduled for 2026.

Computer Vision Module for Termite Detection — IFTM

Apr 2026 - Dec 2026

YOLO · Python · Drone Multispectral

- Developing a module for automated identification and quantification of termite infestations via multispectral drones, integrating multispectral image analysis with object detection.

Seed Germination Classification — IFTM

Apr 2023 - Dec 2024

Python · CNN · TensorFlow/PyTorch

- Developed an application with a CNN model for automated seed germination classification, achieving 86% accuracy and reducing analysis time from ~30s (manual process) to 8s per batch — an efficiency gain of ~73%.
- **Registered at the INPI under No. BR512024005247-0.**

SIPADE Project — IFTM

Jul 2025 - Jul 2026

Docker · GitHub Actions · Next.js · PostgreSQL · Express

- Refactored and expanded the mobile application with ~10 new flows and 20+ revised screens, alongside delivering 10 new features.
- Developed 2 screens in the web application (Next.js, PostgreSQL, Express) and implemented the Offline-First pattern, ensuring continuous field operation without connectivity and automatic synchronization upon reconnection.
- Automated the CI/CD pipeline and containerized microservices with Docker, optimizing orchestration and deployment of georeferencing solutions.

EDUCATION

Instituto Federal do Triângulo Mineiro

Uberlândia, MG

B.S. in Computer Engineering (9th Term)

Mar 2022 - Dec 2026 (Expected)

- Expected completion of all coursework by July 2026.
- Teaching Assistant in Probability and Statistics (2026): probability, random variables, estimation, and hypothesis testing.

CERTIFICATIONS & AWARDS

INPI Registry No. BR512024005247-0 — Seed Germination Classification Software. 2024

1st Place — HackaGIS Health. 2023

3rd Place — IFTM Hackathon. 2024

SKILLS

Programming: TypeScript/JavaScript, Python, Dart, C#, SQL

Front-end: React, Angular, Next.js, Tanstack

Back-end: NestJS, Express, FastAPI, .NET

Mobile: Flutter

Data & AI: TensorFlow, PyTorch, Jupyter, YOLO

Databases: PostgreSQL, MySQL/MariaDB, MongoDB, Qdrant

DevOps & Cloud: Docker, Git, GitHub Actions, Firebase, Azure, AWS

Languages: Portuguese (Native), English (Upper-Intermediate)